Karim Naguib

Economist/Data Scientist | Bayesian Methods & Causal Inference

karim.a.naguib@gmail.com | karimn.github.io | github.com/karimn

Summary

Economist and applied Bayesian statistician with 10+ years building **hierarchical models** for **longitudinal and survival data**. Current focus: **patient-level digital-twin models** of oncology tumor dynamics and real-world evidence integration at AstraZeneca; prior work spans field experiments in East Africa and South Asia, real-estate market design, and health-behavior interventions.

Skills

Methods: Bayesian hierarchical modeling, state-space and longitudinal models, multistate survival and competing risks, joint longitudinal–survival ("digital twin") models, cross-validation and predictive assessment (LFO-CV, PSIS-LOO), MCMC diagnostics and prior sensitivity, causal inference and randomized experiment design, Gaussian processes

Languages: R (tidyverse), Stan, Julia, SQL, C/C++

Agentic Workflows for Statistical Modeling: Built and maintain an agentic-workflow layer on Claude Code, shared across the AstraZeneca modeling team via an internal plugin marketplace. Collaborative mathematical derivation, Stan diagnostic sub-agents, `targets` pipeline explorer, MCP-integrated skills for Domino, reusable plugin and skill templates codifying team conventions for Bayesian workflow.

Infrastructure: CmdStan / cmdstanr, `targets` pipelines, Domino, Docker, AWS, Databricks, Snowflake, PostgreSQL, git, SLURM, RStudio / VS Code

Education

Boston University | Boston, MA | **Ph.D. in Economics** | 2014 *Fields: development, health, and experimental economics*

The American University in Cairo | Cairo, Egypt | **B.S. in Computer Science, Minor: Mathematics** | 1999

Experience

Senior Data Scientist | AstraZeneca | Aug 2023 – present

- Bayesian digital-twin modeling of oncology tumor dynamics to forecast Phase 3 outcomes from Phase 2 trials, historical trials, and real-world data, supporting Phase 3 initiation decisions
- Built multistate (illness-death) survival models and hierarchical state-space tumor-dynamics models, jointly fit for competing-risk decomposition of progression-free survival
- Designed and shared an agentic-workflow layer for collaborative mathematical derivation and MCMC diagnostics, adopted across the modeling team
- Contributed modular Stan architecture and a `targets`-based pipeline for reproducible trial analyses

Senior Data Scientist | Opendoor | May 2021 – May 2022

- Designed randomized experiments to estimate demand and supply elasticities for pricing policy
- Built hierarchical Bayesian and Gaussian-process models pooling across regional real-estate markets

Independent Researcher / Consultant | May 2019 – May 2021

- Built structural Bayesian models of social-experiment outcomes; hierarchical intervention-effect models
- Estimated time-varying COVID-19 transmission dynamics during the early pandemic

Economist | Evidence Action | May 2014 – May 2019

- Designed and analyzed large-scale randomized evaluations in Kenya and Bangladesh
- Contributed to the at-scale replication analysis of No Lean Season (a seasonal-migration subsidy program); the failure to replicate the pilot effect informed the program's eventual closure
- Spanned study design, power analysis, field data-collection supervision, and Bayesian/frequentist analysis for operational and policy audiences

Software Design Engineer in Test | Microsoft, Windows Debugging Tools | Apr 1999 – Mar 2007

- Test framework and automation for Windows debugging tools; C/C++, systems-level testing

Publications

Barker, Davis, López-Peña, Mitchell, Mobarak, Naguib, Reimão, Shenoy, and Vernot, "Migration and Resilience during a Global Crisis," *European Economic Review* (accepted); two additional working papers on migration at scale (Bangladesh) and optimal incentives under social norms (Kenya).